

HACKERRANK ALGORITHMIC PROBLEM-SOLVING & PORTFOLIO INTEGRATION

3rd Semester CSE Portfolio

Name: [PRIYA K]

USN / Student ID: R25EJ110

Course: B.Tech Computer Science & Engineering

Semester: 3rd Semester

Programming Language: Python 3

HackerRank Profile: [priyu_6363 | HackerRank](#)

GitHub Repository: [HackerRank-3rdSem-Portfolio/README.md at main · PriyaK196/HackerRank-3rdSem-Portfolio](#)

1. Objective

The objective of this activity was to strengthen algorithmic problem-solving skills by solving programming problems on HackerRank and documenting the solutions in a structured GitHub portfolio. The activity focused on understanding algorithms, data structures, computational complexity, and practical use of version control.

The solutions were implemented using Python 3 and tested on HackerRank before being organized and documented in a public GitHub repository.

2. Problems Solved

No.	HackerRank Problem	Main Concept	Time Complexity	Space Complexity
1	Diagonal Difference	Matrix Traversal	$O(N)$	$O(1)$
2	Dynamic Array	Dynamic Arrays	$O(N + Q)$	$O(N)$
3	Time Conversion	String Processing	$O(1)$	$O(1)$
4	Compare the Triplets	Conditional Logic	$O(1)$	$O(1)$
5	Sparse Arrays	Hash Maps	$O(N + Q)$	$O(N)$

3. Problem Descriptions and Solutions

3.1 Diagonal Difference

The Diagonal Difference problem requires calculating the absolute difference between the sums of the primary and secondary diagonals of a square matrix. The solution traverses the matrix once. The primary diagonal elements are accessed using `arr[i][i]`, while the secondary diagonal elements are accessed using `arr[i][n-i-1]`

Time Complexity: $O(N)$

Space Complexity: $O(1)$

3.2 Dynamic Array

The Dynamic Array problem involves maintaining multiple sequences and processing two types of queries. The sequence used for each query is determined using XOR between the given value and `lastAnswer`. The solution uses a list of lists to store the sequences and updates `lastAnswer` whenever a type-2 query is processed.

Time Complexity: $O(N + Q)$

Space Complexity: $O(N)$

3.3 Time Conversion

The Time Conversion problem converts a time from 12-hour AM/PM format into 24-hour format. The solution extracts the hour and AM/PM indicator and handles the special cases of 12 AM and 12

Time Complexity: $O(1)$

Space Complexity: $O(1)$

3.4 Compare the Triplets

The Compare the Triplets problem compares three scores belonging to two participants. Each corresponding pair of scores is compared, and a point is awarded to the participant with the higher score.

Time Complexity: $O(1)$

Space Complexity: $O(1)$

3.5 Sparse Arrays

The Sparse Arrays problem requires counting how many times each query string occurs in a given collection of strings. A Python dictionary is used to store the frequency of each string. The dictionary allows the query results to be obtained efficiently.

Time Complexity: $O(N + Q)$
Space Complexity: $O(N)$

4. Methodology: The activity was completed through the following process:

1. The assigned HackerRank problems were identified and opened.
2. Python 3 was selected as the programming language.
3. Each problem was analyzed to understand its input, output, and constraints.
4. An appropriate algorithm and data structure were selected.
5. The solution was implemented in Python.
6. The code was tested using HackerRank test cases.
7. Successfully submitted solutions were documented using screenshots.
8. The source code was organized into separate folders in a GitHub repository.
9. A README file was created to document the problems, concepts, complexity, and learning outcomes.
10. The HackerRank Problem Solving profile was progressed to the required **3-Star** level.

5. GitHub Portfolio Structure: The solutions were organized in the following repository structure:

```
HackerRank-3rdSem-Portfolio/  
├── README.md  
├── Diagonal-Difference/  
│   └── solution.py  
├── Dynamic-Array/  
│   └── solution.py  
├── Time-Conversion/  
│   └── solution.py  
├── Compare-the-Triplets/  
│   └── solution.py  
├── Sparse-Arrays/  
│   └── solution.py  
└── screenshots/  
    ├── problem-solving-3-star.png  
    ├── diagonal-difference.png  
    ├── dynamic-array.png  
    ├── time-conversion.png  
    ├── compare-triplets.png  
    └── sparse-arrays.png
```

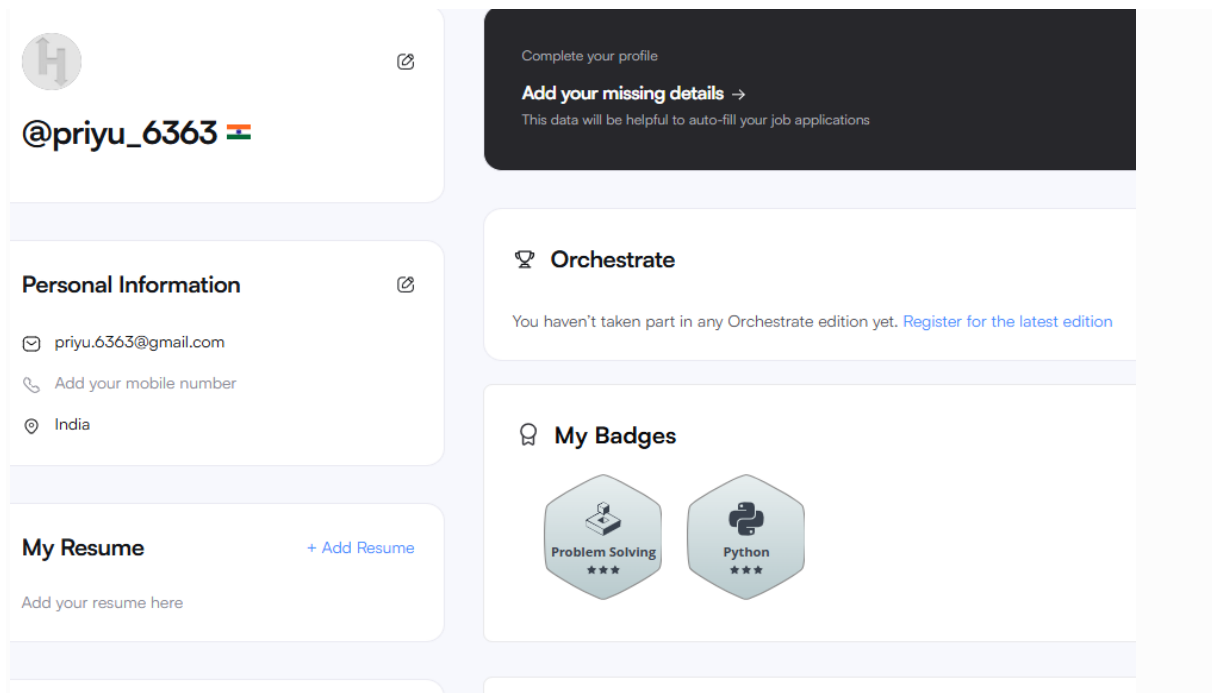
GitHub Repository:

[PriyaK196/HackerRank-3rdSem-Portfolio](https://github.com/PriyaK196/HackerRank-3rdSem-Portfolio)

6. HackerRank Achievement

The HackerRank Problem Solving profile was progressed from the initial level to the required **3-Star** level.

Achievement Evidence:



The screenshot demonstrates the achieved Problem Solving star level and provides evidence of the HackerRank progress.

7. Learning Outcomes:

This activity strengthened my understanding of matrix traversal, dynamic arrays, string processing, conditional logic, and hash maps. I also practiced analyzing time and space complexity and learned to organize programming solutions and evidence using GitHub..

8. Reflection

The activity improved my confidence in solving programming problems using Python. I learned to select suitable data structures, analyze algorithm efficiency, and handle HackerRank function templates and test cases. Creating the GitHub portfolio also gave me practical experience with repository organization, documentation, and version control.

9. Conclusion

The activity combined HackerRank problem solving with GitHub portfolio development. The completed solutions, complexity analysis, screenshots, and 3-Star achievement provide evidence of my algorithmic problem-solving practice and technical documentation skills.

GitHub Repository: [PriyaK196](#)

HackerRank Profile: [priyu_6363 | HackerRank](#)